

Generation Interconnection System Impact Study Report

For Queue Project Q17-0854

Fair Oaks 138 kV

August 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0854. The Interconnection Customer proposes a 103.1 MW solar photovoltaic facility interconnecting at the Fair Oaks 138 kV substation in the Pennsylvania area. The request was received on August 28, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 260 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,166,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wyoming / North Lewiston 138 KV	West Waynesboro / Chambersburg 345 KV	127.3	441	5.8
East Devon 345/138 kV Autotransformer	Fair Oaks / Chambersburg 345 KV Line	123.9	300	7.6
N Lewiston / Chambersburg 138 KV	W Waynesboro / Chambersburg 345 KV	122.3	492	8.1
Otsgo / M105 Substation 138 KV L	W Waynesboro / Chambersburg Substation 345 KV	118.7	471	5.8
Chambersburg 345 kV/138 kV Transformer Bank	West Waynesboro / Chambersbrg 345 KV	116.6	644	10.1
Carrel / Lourich Substation 138 KV line	Fair Oaks / Chambersburg 345 KV L	111.5	376	9.8
Antrim / Peterson Flats 138 KV	Fair Oaks / Chambersburg 345 KV Line	107.1	297	5.2
Peterson Flats / Otsego 138 KV	Fair Oaks / Chambersburg Substation 345 KV	103.6	291	5.2
Carrel 345-138 kV Transformer	Fair Oaks / Chambersburg 345 KV Line	102.7	1,121	29.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wyoming / N Lewiston Substation 138 KV	Rebuild line with higher-capacity conductor	\$1,040,000	3% (\$33,000)
East Devon 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$11,815,000	5% (\$638,000)
N Lewiston / Chambersburg 138 KV	Rebuild line with higher-capacity conductor	\$4,559,000	4% (\$199,000)
Otsgo / M105 Substation 138 KV L	Rebuild line with higher-capacity conductor	\$5,978,000	3% (\$191,000)
Chambersburg 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,128,000	4% (\$695,000)
Lourich / Carrel 138 KV	Rebuild line with higher-capacity conductor	\$7,508,000	5% (\$355,000)
Peterson Flats / Antrim 138 KV	Reconductor line	\$1,570,000	5% (\$77,000)
Peterson Flats / Otsego 138 KV	Reconductor line	\$1,674,000	5% (\$82,000)
Carrel 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$18,699,000	5% (\$896,000)
	Total allocated to project		\$3,166,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.